

JARED DUONG

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📁 [Portfolio](#)

Experience

Software Integration Developer

Jan 2026 – Mar 2026

Appleby North Paint Inc.

Remote

- Built a **REST-based integration** between **QuickBooks** and **Decor Fusion** to automate sales and inventory data synchronization.
- Mapped invoice and transaction data and built **API-based workflows** to reduce manual data entry.
- Created technical documentation explaining integration setup, data flow, and maintenance procedures for future updates.

Production Student

May 2025 – Jun 2025

Hartmann Canada

Brantford, ON

- Worked with automated manufacturing equipment and reviewed production data to identify recurring process issues.
- Helped troubleshoot mechanical and production problems alongside operators and engineers in a manufacturing environment.
- Documented recurring equipment and system issues to support production monitoring and process improvements.

Projects

16-Bit ALU Design and FPGA Implementation | VHDL, FPGA, Digital Logic

Jan 2026 – Feb 2026

- Designed modular 1-bit ALU slices and composed a scalable 16-bit ripple-carry architecture in **VHDL**.
- Implemented arithmetic and logic operations including add, subtract, **AND**, **OR**, **XOR**, and **NOT**.
- Developed a **VHDL** testbench to simulate ALU behavior across multiple input combinations.
- Validated behavior on **Nexys-3 FPGA** hardware using simulation and debugging tools.

Reverse Engineering & CAD Reconstruction | SOLIDWORKS

Jan 2026 – Feb 2026

- Reverse engineered a mechanical assembly in **SOLIDWORKS**, modeling its components and original geometry.
- Reconstructed component placement and mechanical relationships within the CAD assembly.
- Analyzed component interactions and mechanical constraints to understand how the assembly functions.

4x4x4 LED Cube with Custom PCB | Arduino, PCB Design, Digital Logic

Mar 2025 – Apr 2025

- Designed and soldered a 64-LED cube using discrete logic ICs (**74HC04**, **SN74HC08**) and a custom-routed **PCB**.
- Programmed multiplexed LED control patterns using an **Arduino Nano** in **C/C++**.
- Implemented layered scanning logic to control 3D LED addressing and timing.
- Designed and 3D-printed a structural base to house power and control electronics.

Autonomous Wheelchair Prototype | Arduino, C++, SOLIDWORKS

Sep 2024 – Dec 2024

- Developed an **Arduino**-powered autonomous wheelchair prototype integrating sensors and motor control for navigation.
- Performed **center-of-mass and tipping analysis** to guide mechanical redesign and improve prototype stability.
- Evaluated component costs and optimized the prototype design to reduce expenses.

Education

University of Guelph

Sep 2024 – Apr 2028

Bachelor of Engineering in Computer Engineering (Co-op)

Guelph, ON

Technical Skills

Languages: C, C++, VHDL, Python, Java, SQL, C#

Embedded / Hardware: Arduino, FPGA (Nexys-3), Digital Logic Design, Sensor Integration, Motor Control, PCB Design, Soldering, Circuit Analysis, Robotics

Tools: Vivado, ModelSim, MATLAB, Git & GitHub, Visual Studio Code

CAD / Design: SOLIDWORKS (Certified SOLIDWORKS Associate, CSWA), Fusion 360, AutoCAD, 3D Printing

Leadership / Extracurricular

Robotics Club – Mechanical Team Member

Sep 2024 – Present

University of Guelph

Guelph, ON

- Design and build stable mechanical components for competition robots.
- Work with electrical and software teammates to integrate sensors, controls, and mechanical components.
- Model and prototype parts in **SOLIDWORKS** and **AutoCAD**, test designs, and resolve assembly issues.